Pharmaceutical Molecular Design Agent (PMDA): A Governed Multi-Agent Framework for Target-Specific Drug Candidate Generation and Evaluation

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in Partial Fulfillment of the Requirements
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Bachelor of Technology

by

Harsh Shukla
(Roll No. 230122018)

under the guidance of

Dr. Kalishankar Bhattacharyya



to the

DEPARTMENT OF CHEMISTRY
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI
GUWAHATI–781039, ASSAM, INDIA

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Declaration

I declare that the work presented in this project report entitled **"Pharmaceutical Molecular Design Agent (PMDA): A Governed Multi-Agent Framework for Target-Specific Drug Candidate Generation and Evaluation,"** submitted to the Department of Chemistry, Indian Institute of Technology Guwahati, under the guidance of **Dr. Kalishankar Bhattacharyya**, in partial fulfillment of the requirements for the degree of Bachelor of Technology, is my original work. I further declare that this work has not been plagiarized or previously submitted, either in full or in part, to this or any other institution for the award of any degree, diploma, or academic qualification. All research articles, datasets, software, computational resources, and other external materials used in this project have been appropriately acknowledged and cited.

Date: _____ Harsh Shukla

Roll No. 230122018

Signature: _____

Certificate

This is to certify that the work contained in the project report entitled **"Pharmaceutical Molecular Design Agent (PMDA): A Governed Multi-Agent Framework for Target-Specific Drug Candidate Generation and Evaluation"** is the bona fide work of **Harsh Shukla (Roll No. 230122018)**, carried out in the **Department of Chemistry, Indian Institute of Technology Guwahati**, under my supervision, and that this work has not been submitted elsewhere for the award of any degree, diploma, or other academic qualification.

Guwahati
Date: _____

Dr. Kalishankar Bhattacharyya
Project Supervisor

Signature: _____

Acknowledgements

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Executive Summary

This project presents the **Pharmaceutical Molecular Design Agent (PMDA)**, a governed multi-agent framework for target-specific drug candidate generation and computational evaluation. PMDA combines four specialist agents - Planner, Formulator, Executor Assistant, and Analyser - with a deterministic controller that manages approvals, execution limits, campaign states, and evidence records.

Human wild-type **EGFR** was selected as the initial target. From 18,381 public assay records, 777 molecules were curated and divided into training, calibration, and test sets. On the scaffold-held-out test set, the predictor obtained an R2 score of 0.6432 and a mean absolute error of 0.5573 pIC50. Performance improved for molecules within the model's applicability domain, supporting its use as a preliminary ranking model rather than a substitute for experimental testing. Prediction intervals and applicability-domain checks were used to identify uncertain or unfamiliar molecules.

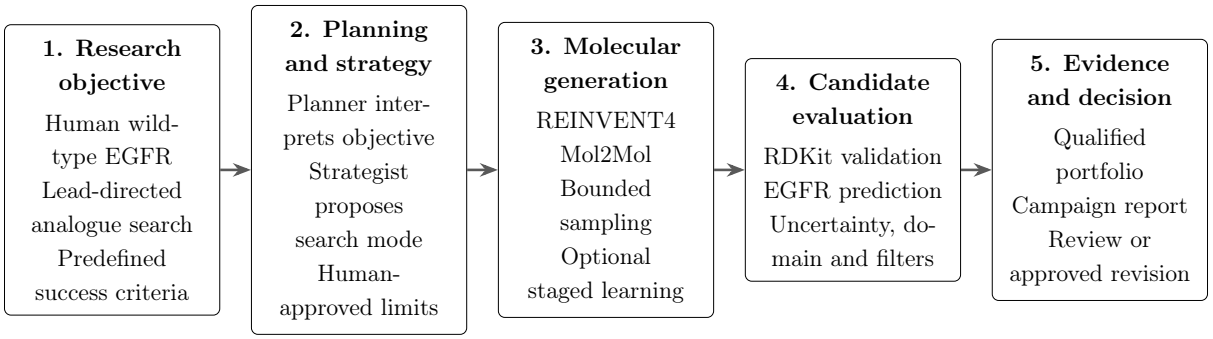
The complete campaign generated and evaluated **112 molecular structures** across three controlled jobs. Three molecules satisfied all predefined computational criteria, with predicted pIC50 values of **7.815, 7.588, and 7.470**. A separate benchmark showed that the multi-agent workflow improved traceability and control, although it did not demonstrate better molecular outcomes than the fixed workflow.

PMDA therefore provides an integrated and auditable research prototype for molecular generation, prediction, evaluation, and human review. Its selected structures are computationally prioritized drug candidates that require further experimental validation.

Project Workflow at a Glance

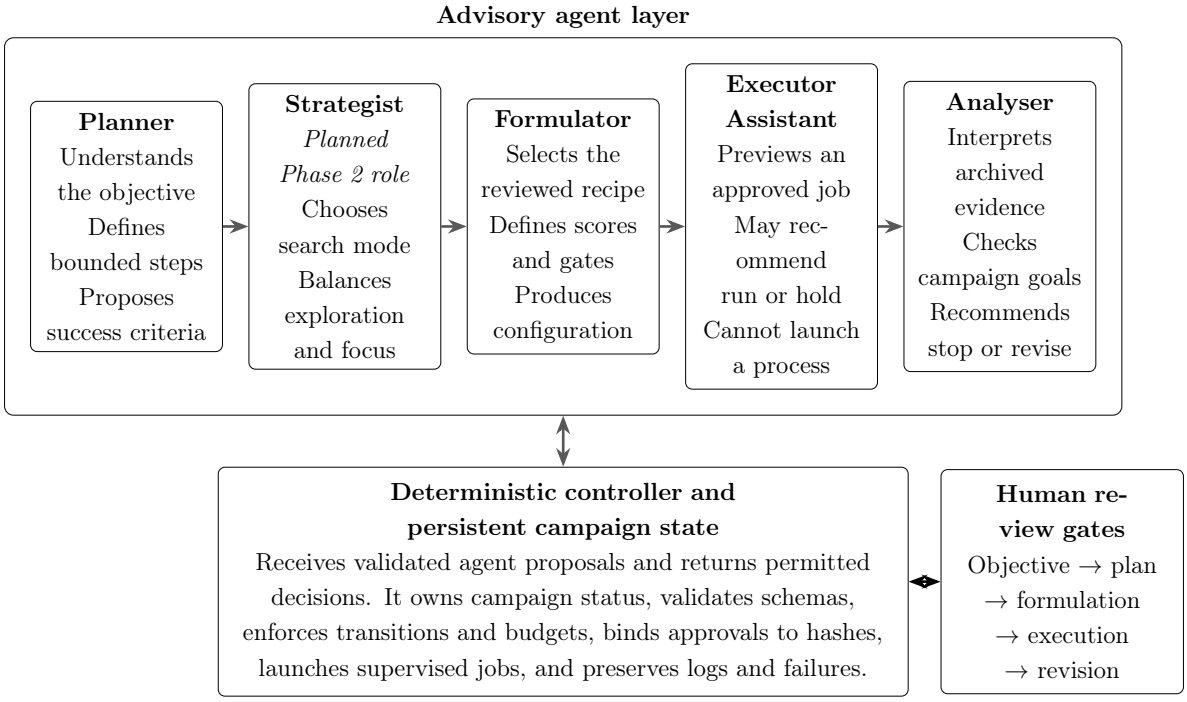
The following three views introduce the project before the detailed chapters. They distinguish the complete research pipeline, the internal agent orchestration, and the observed EGFR campaign. Solid arrows represent executed information flow; review gates remain under deterministic controller and human authority.

Workflow 1: End-to-End PMDA Research Pipeline



Example. A scientist requests analogues around a registered EGFR lead. The campaign does not return every generated SMILES as a result. Structures first pass chemical validation, target-specific prediction, empirical uncertainty, applicability-domain and policy gates. Only candidates satisfying the frozen portfolio conditions are reported as computational priorities.

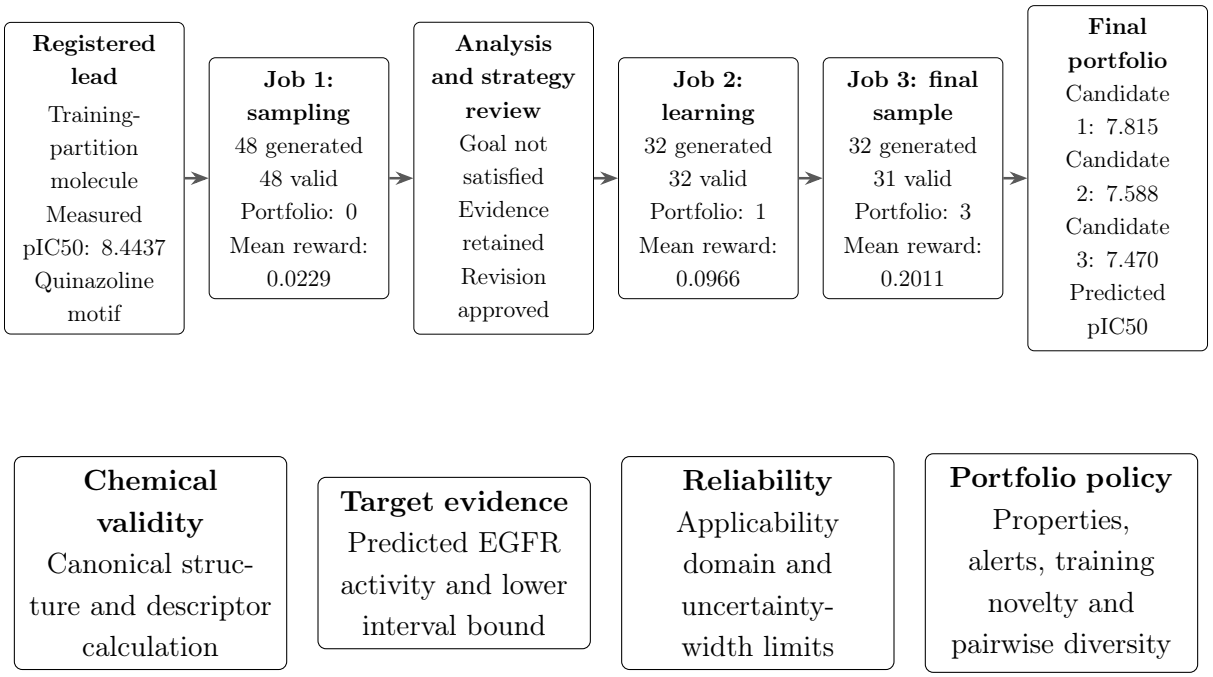
Workflow 2: Governed Multi-Agent Orchestration



The language-model roles are advisory. Execution authority, state changes and resource limits belong to the controller. Planner, Formulator, Executor Assistant and Analyser are implemented. The Strategist is shown to communicate the intended complete architecture, but remains a planned Phase 2 role.

Worked Example: Adaptive EGFR Campaign

Workflow 3: From Lead Molecule to Final Computational Portfolio



How to read this example. The campaign evaluated 112 generated rows across three controlled jobs; 111 rows were chemically valid. Increasing mean reward is an observed campaign trajectory, not proof of experimental potency improvement. The three final structures passed the predefined computational gates, but they remain hypotheses requiring medicinal-chemistry assessment, synthesis and biological testing.

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1 Introduction

1.1 Background: Drug Discovery and Molecular Design

The discovery of a new drug begins with a biological question but quickly becomes a search problem. A useful small molecule must interact with the intended biological target while also satisfying structural, physicochemical, safety and developability requirements. These requirements can conflict: a change that improves predicted target activity may increase lipophilicity, reduce solubility or move the molecule outside the region represented by available experimental data. Drug design is therefore a multi-objective process rather than the optimization of a single score.

Traditional discovery programmes reduce this search space through medicinal-chemistry knowledge, experimental screening and repeated design–make–test–analyse cycles. Computational methods can support the design stage by representing molecules numerically, learning relationships from measured compounds and prioritizing a smaller set for further study. Their value depends on the quality of the underlying assays and on whether a proposed molecule resembles the chemistry on which a model was trained.

1.2 The Challenge of Chemical-Space Exploration

The number of chemically possible drug-like structures is far larger than any library that can be synthesized or tested. Exhaustive enumeration is consequently impractical. A molecular generator can explore this space more selectively, but unrestricted generation creates a second problem: syntactically valid molecules may still be irrelevant to the chosen target, chemically repetitive, outside the predictor's applicability domain or unsuitable for follow-up.

This distinction between *generation* and *qualification* is central to the project. A generated SMILES string is only a proposal. It becomes a computational candidate only after structure validation, property calculation, target-specific prediction, uncertainty assessment, domain checking and policy review. Even after these steps, it remains a hypothesis rather than an experimentally established drug candidate.

1.3 Generative Molecular Design

Modern molecular generators learn patterns from known chemical structures and use them to sample new structures. Lead-oriented generation can focus the search around a reference molecule, making it suitable for analogue exploration. In this project, REINVENT4 provides the Mol2Mol generation engine [2]. RDKit is used to parse, standardize and calculate chemical representations and properties [3].

Generation and prediction must not be treated as independent black boxes. If a predictor supplies the reward used during generation, prediction errors can direct the search toward confidently incorrect regions. PMDA therefore combines prediction with empirical uncertainty intervals, similarity-based applicability-domain checks and fixed

acceptance gates. The objective is controlled prioritization, not a claim of autonomous drug discovery.

1.4 Agentic Workflows and Scientific Control

Multi-agent systems can divide a complex task into specialist responsibilities. Solitarius demonstrated this idea for computer-aided molecular inverse design by separating planning, formulation, execution and analysis roles [1]. Such separation can make a workflow easier to inspect, but an agent response alone should not authorize scientific execution. Language-model outputs may be incomplete, inconsistent or unsupported by the available evidence.

PMDA adopts the role-separation concept while implementing an independent orchestration layer. A deterministic controller owns campaign state, approvals, budgets, file hashes and process launch. Specialist agents produce bounded proposals through schemas and restricted context. This arrangement creates a visible boundary between advisory reasoning and executable action.

1.5 Problem Statement

Available molecular-design components often operate as separate tools: a generator proposes structures, a predictor assigns activity values, scripts apply filters and results are inspected manually. Without a shared campaign record, it becomes difficult to determine which objective, dataset, model, configuration and approval produced a particular molecule. It is also easy to report a high predicted score without displaying uncertainty, applicability limits or failed candidates.

The problem addressed in this work is therefore to construct a local, reproducible and reviewable system that connects target-specific molecular generation with conservative computational evaluation, while preserving the complete chain from research objective to final evidence.

1.6 Objectives

The principal objective is to develop and evaluate a governed multi-agent framework for target-specific drug candidate generation. The specific objectives are:

1. to curate a reproducible EGFR activity dataset and train a scaffold-evaluated prediction model;

2. to integrate lead-oriented molecular generation with structure validation, property calculation, uncertainty estimation and applicability-domain assessment;

3. to implement Planner, Formulator, Executor Assistant and Analyser roles under deterministic controller authority;

4. to introduce explicit approval gates, execution budgets, state transitions and evidence-preserving campaign archives;
5. to select a small portfolio using fixed activity, uncertainty, novelty and diversity criteria;
6. to expose the workflow through a local research portal; and
7. to compare the complete agent workflow with simpler generation and fixed-control alternatives.

1.7 Scope, Contributions and Attribution

The demonstrated scientific scope is lead-oriented generation for human wild-type EGFR using public activity records and local CPU execution. The work does not establish clinical efficacy, selectivity, toxicity, synthetic accessibility at experimental depth or prospective biological activity. The reported candidates are computationally prioritized structures requiring expert and experimental validation.

Solitarius supplies the conceptual role separation and orchestration framework [1]. The unavailable orchestration source was not reproduced at source level. PMDA implements its own controller, schemas, prompts, context restrictions and recovery behavior. REINVENT4 supplies generation [2], RDKit supplies chemical processing [3], and ChEMBL supplies assay records [4]. These contributions remain distinct from project-authored infrastructure.

2 Theoretical Background and Literature Review

2.1 Computer-Aided Drug Design

Computer-aided drug design uses structural and experimental information to prioritize compounds before costly laboratory work. Ligand-based methods infer patterns from known molecules, whereas structure-based methods use a three-dimensional target model. PMDA's demonstrated workflow is primarily ligand-based: measured EGFR inhibitors support the activity predictor, and a known active training molecule guides analogue generation. No docking score is used as evidence in the reported campaign.

The literature and software foundation of the project combines public bioactivity data from ChEMBL [4], cheminformatics through RDKit [3], generation through REINVENT4 [2], and an orchestration concept inspired by Solitarius [1]. PMDA connects these components through its own validation, approval and evidence layers.

2.2 Molecular Representation and Similarity

SMILES strings provide the interchange representation. Canonicalization supports consistent comparison of valid structures. Morgan fingerprints encode local environments

into 2048 bits using radius two. For binary fingerprints a, b, Tanimoto similarity is

$$T(a,b) = \frac{|a \cap b|}{|a| + |b| - |a \cap b|}. \tag{1}$$

Similarity measures proximity to training structures and redundancy within portfolios. Exact training-set novelty does not establish global or patent novelty.

2.3 Target Activity and the pIC50 Endpoint

$$\mathrm{pIC}_{50} = -\log_{10}\left(\frac{\mathrm{IC}_{50}}{1\ \mathrm{mol\,L^{-1}}}\right) = 9 - \log_{10}\left(\frac{\mathrm{IC}_{50}}{1\ \mathrm{nmol\,L^{-1}}}\right). \tag{2}$$

The logarithmic transformation accommodates activity values spanning several orders of magnitude. Unit consistency and assay identity must be established before aggregation.

2.4 Lead-Oriented Molecular Generation

REINVENT4 supports Mol2Mol generation, in which a molecular representation is conditioned on a starting structure and sampled to propose related chemistry [2]. This approach was selected because the project concerns controlled lead exploration rather than unrestricted de novo sampling. Sampling temperature, row limits and execution time are bounded by the campaign configuration. A later staged-learning step can update the generative policy using the configured reward, but its output is evaluated using the same independent gates as ordinary samples.

2.5 Random-Forest Activity Prediction

A random forest averages tree predictions:

$$\widehat{y}(x) = \frac{1}{B}\sum_{b=1}^{B} f_b(x). \tag{3}$$

The implemented model uses 200 trees, minimum leaf size two, feature fraction 0.5 and seed 42. Fingerprints provide the inputs. Performance is evaluated on scaffold-held-out molecules.

2.6 Uncertainty Calibration

Calibration residuals are $r_i = |y_i - \widehat{y}(x_i)|$. The exact implementation uses

$$p = \min\left(1, \frac{\lceil (n_{\mathrm{cal}} + 1)0.9 \rceil}{n_{\mathrm{cal}}}\right), \qquad q = Q_{\mathrm{higher}}(\{r_i\}, p), \tag{4}$$

and reports $[\widehat{y}(x) - q, \widehat{y}(x) + q]$. Empirical held-out coverage is measured separately; coverage for generated molecules under distribution shift is not guaranteed.

2.7 Applicability Domain

Nearest-training similarity is

$$s(x) = \max_{z \in \mathcal{D}_{\text{train}}} T(x, z). \tag{5}$$

The domain threshold is the larger of 0.35 and the fifth percentile of nearest-other-training similarities. A molecule is in domain when $s(x)$ reaches this threshold.

2.8 Conservative Reward and Portfolio Selection

Let $g(x)$ indicate passage through every configured gate, and let $L(x) = \widehat{y}(x) - q$. The implemented reward is

$$R(x) = g(x) \min\left(1, \max\left(0, \frac{L(x) - 5}{3}\right)\right). \tag{6}$$

The lower bound reduces reward for uncertain predictions. This remains a prioritization criterion rather than a probability of experimental success.

The selected portfolio P must contain at least three molecules satisfying $g(x) = 1$, $L(x) \geq 6$, absence from the training structures and pairwise similarity $T(x_i, x_j) \leq 0.85$. Thresholds are fixed before sampling.

2.9 Solitarius-Inspired Multi-Agent Orchestration

Solitarius motivates the separation of a molecular-design task into planning, formulation, execution and analysis stages [1]. PMDA retains these functional roles but does not attempt to recreate unavailable source code or prompts. Its controller is project-authored and treats agent messages as structured recommendations. State transitions, authorization, resource limits and execution remain deterministic operations. This distinction permits the scientific role of each agent to be studied without granting an unconstrained language model direct control of computational jobs.

2.10 Evaluation Metrics

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \widehat{y}_i|, \tag{7}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \widehat{y}_i)^2}, \tag{8}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{9}$$

Interval coverage measures the fraction inside reported intervals. Domain coverage measures the fraction passing the similarity threshold. Campaign metrics include generated rows, validity, portfolio size, runtime and failures.

3 System Design and Methodology

3.1 Overall Research Workflow

The work proceeds through six linked stages: target and endpoint registration; assay curation; predictor training and qualification; lead and campaign definition; governed molecular generation; and candidate evaluation with evidence preservation. Scientific artefacts move forward only after validation. This prevents a later campaign from silently changing the dataset, model or acceptance criteria used to justify its results.

3.2 EGFR Target and Dataset Preparation

3.2.1 Target Selection

EGFR was selected as the first demonstrator because it is a pharmaceutically relevant kinase target with sufficient public small-molecule activity records to support a student-scale modelling study. The registered entity is human wild-type EGFR, identified as CHEMBL203 and UniProt P00533. Restricting the target identity avoids mixing measurements from unrelated proteins or unspecified target variants.

3.2.2 Assay Acquisition and Curation

The registered target is human EGFR, CHEMBL203/P00533. The frozen collection contains 18,381 exact IC50 binding records. Conservative target and assay screening, structure processing and duplicate aggregation produced 777 molecules and 354 scaffold categories. Row-level curation decisions are preserved.

3.2.3 Scaffold-Based Data Partitioning

Scaffold groups are ordered using a seeded SHA-256 rule independent of endpoint labels. The first 65% of groups enter training, the next 15% calibration and the remainder testing. Group allocation yields 518, 88 and 171 molecules; molecule proportions need not equal scaffold-group proportions.

Table 1: Frozen dataset composition.

Quantity	Count
Source records	18,381
Curated molecules	777
Scaffold categories	354
Training	518
Calibration	88
Test	171

3.3 Predictor Development and Qualification

3.3.1 Fingerprint Construction

Each curated structure is converted to a radius-two, 2048-bit Morgan fingerprint. Identical processing is applied during training and campaign evaluation. Failure to parse or featurize a structure produces a recorded rejection rather than a missing or substituted score.

3.3.2 Model Training

The random-forest hyperparameters are fixed at 200 trees, a minimum leaf size of two, a feature fraction of 0.5 and random seed 42. Training uses only the training partition. The calibration partition determines the empirical interval width, while the scaffold-held-out test partition is reserved for performance reporting.

3.3.3 Uncertainty and Domain Qualification

The 90% empirical residual interval and nearest-training similarity rule described in Section 2 are applied to every valid candidate. Predictions outside the domain remain visible but cannot satisfy the demonstrated portfolio policy.

3.4 PMDA System Architecture

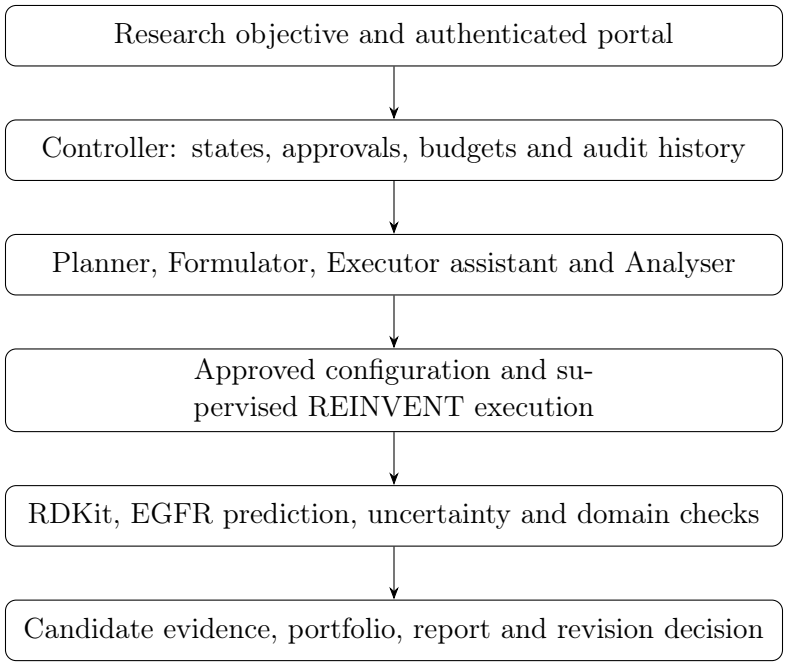


Figure 1: PMDA architecture. Roles advise; the controller retains execution authority.

The controller uses transactional SQLite and explicit transitions. Approvals bind decisions to reviewed artifact hashes. Budget reservations prevent duplicate authorization.

Hash-chained events support detection of modification; an operating-system administrator remains outside this application-level boundary.

3.5 Specialist Agent Responsibilities

Local Qwen2.5-1.5B-Instruct inference runs through llama.cpp. Calls use a response schema, temperature zero, a seed, at most 384 output tokens and two attempts. Context allowlists restrict role-specific inputs.

The Planner proposes a bounded plan. The Formulator selects a reviewed scoring recipe. The Executor assistant proposes a preview or hold and cannot launch processes. The Analyser checks the outcome against execution, budget and portfolio evidence. Rejected replies remain archived. Retries may include controller-supplied routing guidance, limiting claims of independent reasoning.

3.6 Approval, Guardrail and Audit Design

Before execution, the objective, plan, scoring formulation and job preview pass through separate review points. Every approval is bound to the hash of the reviewed artefact. An approved record cannot authorize a subsequently modified configuration. The controller also enforces job count, candidate count, revision and runtime limits. These controls are engineering safeguards; they do not certify molecular safety.

3.7 Lead Selection and Campaign Protocol

The lead is selected from the training partition using activity, motif and property rules. Calibration and test labels do not guide selection. Its SMILES is

```
CCOc1cc2ncnc(Nc3ccc(C)cc3)c2c2c1OCCO2
```

The training-record pIC50 is 8.4437; this assay aggregate is not a measurement of generated analogues.

The first job samples 48 Mol2Mol analogues. If selection is insufficient, an approved revision permits two reinforcement-learning steps of 16 rows each. A second revision samples 32 structures separately from the resulting checkpoint. Limits are three jobs, 112 rows, two revisions and 600 seconds per job. Learning trajectories cannot independently close the campaign.

3.8 Candidate Evaluation Policy

The recipe requires a quinazoline motif, MW at most 500, logP at most five, no configured PAINS/Brenk exclusions, model-domain qualification and interval half-width at most 1.5 pIC50. Additional descriptors include TPSA, donors, acceptors, rotatable bonds, formal charge, QED and a synthesis-accessibility heuristic. Structural alerts are screening flags, not toxicity measurements.

Invalid structures, prediction errors and policy rejections retain reasons. Qualified prediction status differs from portfolio membership, which additionally requires conservative activity, novelty and pairwise similarity conditions.

3.9 Research Portal and Evidence Preservation

The local FastAPI portal supports permissions, separate review gates, candidate inspection, agent records and authenticated downloads. Sessions use hashed tokens, expiry and CSRF checks. The portal runs one bounded sampling job; the adaptive three-job demonstration uses the campaign runner.

Project-specific code, models, data and dependencies reside in BTP. System Windows and Python remain prerequisites. Configurations, inputs, logs and role responses are preserved. Failed or ambiguous jobs do not silently become successful executions.

4 Results and Discussion

4.1 Dataset and Predictor Performance

4.1.1 Overall Test Performance

Table 2: Held-out performance; errors are in pIC50 units.

Stratum	n	MAE	R^2	Coverage
All	171	0.5573	0.6432	0.9357
In domain	136	0.4547	0.7509	0.9779
Outside domain	35	0.9561	-0.4324	0.7714
Measured pIC50 ≤ 5	5	2.1756	-63.5994	0.0000

Overall RMSE is 0.7647 and Spearman correlation is 0.8294. Interval half-width is 1.3983 pIC50. The five weak-activity examples are poorly predicted and have zero interval coverage. Their small number restricts statistical interpretation but identifies a material weakness.

4.1.2 In-Domain and Out-of-Domain Behaviour

The difference between the in-domain MAE of 0.4547 and out-of-domain MAE of 0.9561 shows that similarity to the training chemistry materially affects reliability. The negative out-of-domain R^2 indicates that the model should not be used as a general activity estimator for unfamiliar scaffolds. Domain qualification is therefore used as an acceptance condition rather than displayed only as supplementary information.

4.1.3 Weak-Activity Limitation

Only five test molecules have measured pIC50 at or below five, and their MAE is 2.1756. The model is consequently unsuitable for confidently separating weak or inactive compounds. This imbalance can create optimistic prioritization if high predicted values are interpreted without the test-stratum analysis. Thirty-eight source publications overlap

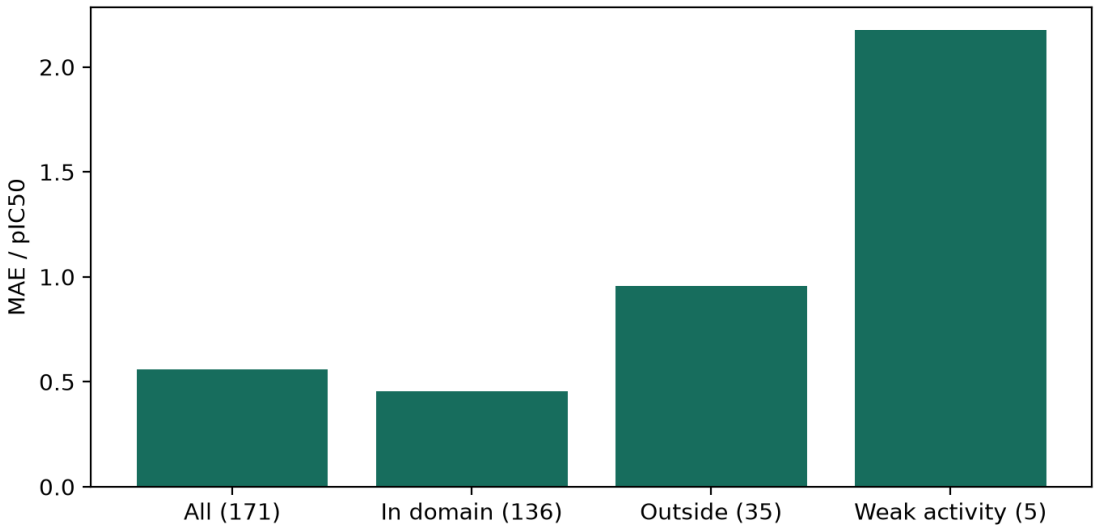


Figure 2: MAE across recorded test strata. Counts appear in parentheses; strata overlap.

across scaffold partitions. Scaffold separation does not remove all source dependence. Assay variability and unknown generator-training overlap further limit generalization.

4.2 End-to-End EGFR Campaign

Table 3: Observed campaign iterations.

Iteration	Mode	Rows	Valid	Portfolio	Mean reward
0	Sampling	48	48	0	0.0229
1	Staged learning	32	32	1	0.0966
2	Final sampling	32	31	3	0.2011

The campaign completed with three final accepted candidates. Across all jobs, 111 of 112 rows were valid; these are not unique-molecule counts. The initial stage failed portfolio requirements. The learning trajectory contained one selected candidate, while final sampling produced three.

Mean reward increased, but the stages differ in mode and sample count. This observation does not establish a causal potency improvement or general optimization advantage.

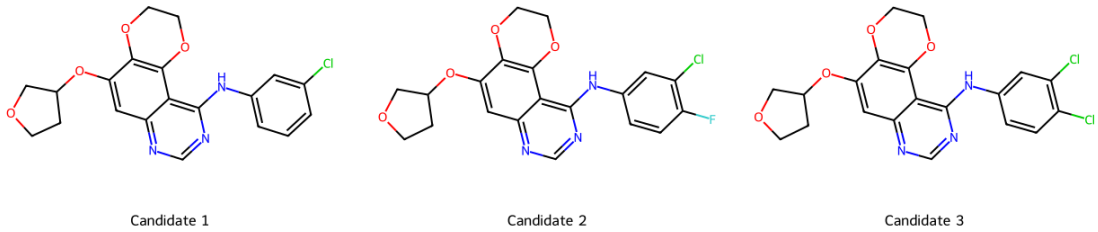


Figure 3: RDKit structures generated directly from archived canonical SMILES.

Table 4: Predicted activity and molecular properties.

Candidate	pIC50	Lower	Upper	MW	logP
1	7.815	6.417	9.214	399.834	3.966
2	7.588	6.190	8.987	417.824	4.105
3	7.470	6.071	8.868	434.279	4.619

4.3 Final Computational Portfolio

All three are in domain, absent from training structures and free from configured alert matches. Each has zero Lipinski violations. They share a scaffold: passing the fingerprint threshold does not establish broad scaffold diversity.

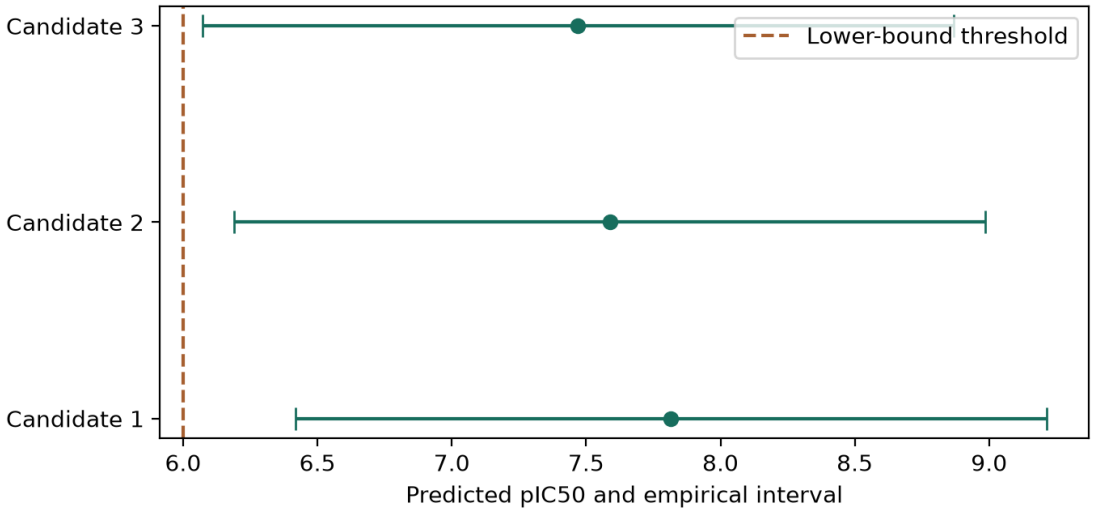


Figure 4: Predictions and empirical intervals. Every lower endpoint exceeds six; the broad intervals retain substantial uncertainty.

4.4 Comparative Workflow Benchmark

Table 5: Portfolio sizes under a 48-row candidate ceiling per arm.

Approach	Seed 42	Seed 99
Prior	0	0
Mol2Mol	0	3
No feedback	0	3
Fixed	2	0
Full agents	2	0

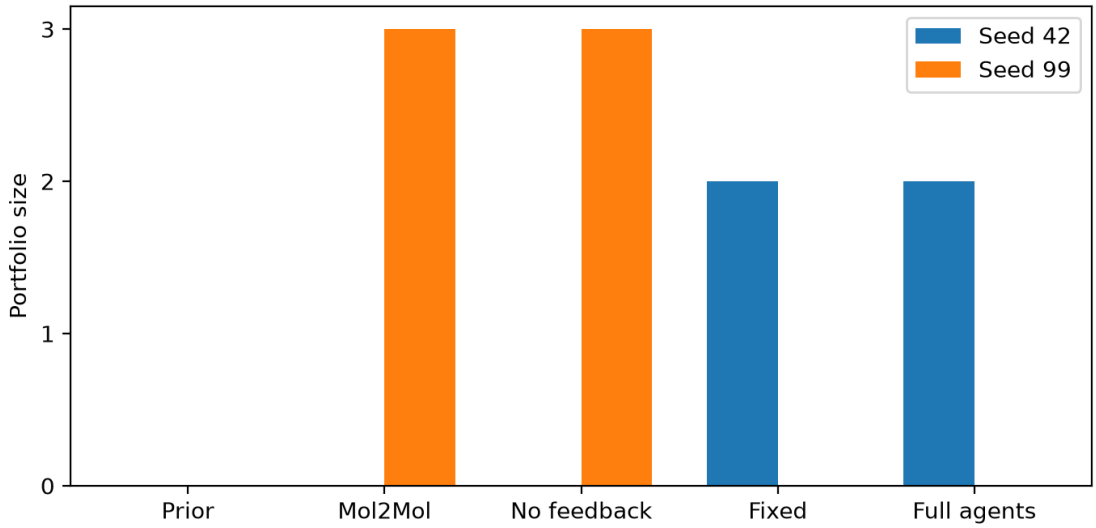


Figure 5: Portfolio sizes in ten pilot runs. Two seeds do not constitute a powered effectiveness study.

The pilot contains 480 generated rows. Full-agent and fixed outputs matched. Adaptive superiority was not established. Candidate ceilings were equal, but training compute and wall time were not. Traceability and review control are engineering outcomes distinct from molecular quality.

4.5 Agent Behaviour and Software Verification

The reference campaign preserved 14 real model attempts, including 12 accepted responses. Seven later diagnostic cases passed final validation: two on their first attempt and five after retry. Controller guidance contributed to retries; the results demonstrate constrained assistance.

The recorded release has 69 passing regression tests without failures or skips. Seven portal tests cover authentication, CSRF, roles, project isolation, approval bypass, download containment and filtering. A real HTTP journey evaluated four molecules through

all approval gates. Pipeline verification passed, but scientific criteria failed and the portfolio was empty. A backup-and-restore drill verified database and audit integrity without launching jobs.

4.6 Scientific and Engineering Limitations

The demonstrated scope is a local CPU prototype and one target. Weak-activity prediction is inadequate and calibration under generated-molecule shift remains unverified. Selectivity, toxicity, synthetic feasibility, global novelty and prospective activity are not established.

Scripted reference approvals represent delegated development authorization, not independent scientific signoff. The portal supports manual reviews, but API checks do not establish complete browser usability. Final visual and accessibility signoff remains pending.

5 Conclusion and Future Scope

5.1 Conclusion

PMDA integrates independent orchestration with molecular generation, target-specific evaluation and auditable evidence. The EGFR campaign demonstrates operation of local constrained agents, deterministic approvals, bounded execution and portfolio selection. Three structures satisfied fixed computational criteria. The comparative pilot did not establish superiority over a fixed workflow.

The outcome is a traceable research prototype with explicit scientific boundaries. Selected structures remain computational hypotheses. Negative outcomes, rejected responses and uncertainty estimates form part of the evidence.

5.2 Future Scope

5.2.1 Strategist and Chemical-Space Intelligence

A Strategist is planned to propose generation modes and exploration policies under approval. Chemical-space intelligence will introduce embeddings, density estimates and regional coverage. VECTOR+ is planned as an optional engine after checkpoint validation and evaluation through a common candidate schema; it is not currently integrated.

5.2.2 Expanded Pharmaceutical Evaluation

Further pharmaceutical scope includes selectivity panels, qualified ADMET endpoints, ensemble uncertainty, Pareto selection and retrosynthesis. Higher-fidelity evaluation may include docking where justified, followed by active learning from new labels. Each evaluator requires independent qualification.

5.2.3 Deployment and Experimental Validation

Engineering priorities include containers, separate web and worker services, quotas, full browser testing and cross-machine recovery. Research priorities include repeated equal-budget comparisons and module ablations. Prospective synthesis and assays require experimental collaboration. New modules should be retained only when measured benefit justifies complexity.

A Evidence Index

Evidence	Location
Campaign root	`runtime/egfr-campaign-20261028T051715Z/`
Lead input	Campaign: `inputs/lead.smi`
Criteria	Campaign: `inputs/criteria.json`
Final samples	Campaign: `outputs/iteration-2/samples.csv`
Evaluated rows	Campaign: `iteration-2/evaluation/candidates.parquet`
Portfolio	Campaign: `final-portfolio.json`
History	Campaign: `TASK_PLAN.md`
Benchmark	`runtime/benchmark-stage8-20261028T052759Z/results.json`
Predictor strata	`runtime/science-stage8-20261029T042609Z/report.json`
Portal journey	`runtime/portal-journey-20261029T042822Z/result.json`
Regression	`runtime/verification/tests.json`

The intermediate reinforcement-learning trajectory should be inspected in its archived job and evaluation outputs, rather than assuming a sampling CSV filename.

B Selected Structures

Candidate 1:

```
Clc1cccc(Nc2ncnc3cc(OC4CCOC4)c4c(c23)OCCO4)c1
```

Candidate 2:

```
Fc1ccc(Nc2ncnc3cc(OC4CCOC4)c4c(c23)OCCO4)cc1Cl
```

Candidate 3:

```
Clc1ccc(Nc2ncnc3cc(OC4CCOC4)c4c(c23)OCCO4)cc1Cl
```

References

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